

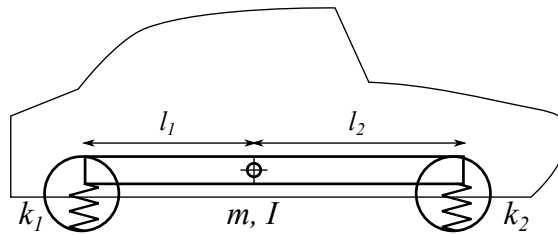
**Example Question 2.**

Figure 2

A car of mass 1600 kg (cf. Fig. 2) has a moment of inertia of  $2300 \text{ kgm}^2$  around its centre of mass, which is located at  $l_2 = 1.70 \text{ m}$  behind the front wheels and  $l_1 = 1.35 \text{ m}$  in front of the rear wheels. The effective spring constants of the rear and front suspension system are  $k_1 = 30.0 \text{ kN/m}$  and  $k_2 = 35.0 \text{ kN/m}$ , respectively.

- i) Select an appropriate coordinate system to describe the vibrations of this system. If possible, avoid dynamic coupling between the coordinates.
- ii) Determine the normal mode frequencies of the car.
- iii) Locate